

Section B:-

Q:2(i)

A:2(i)

Biology:- Biology is the scientific study of life. The word 'biology' is derived from two Greek words. 'Bios' meaning life and 'logos' meaning thought, reasoning and study. Biology deals with all the aspects of living organisms including their growth, structure, function, development, evolution and interactions with the environment.

Three main division of biology:-

*** Botany:-** The scientific study of plants. It includes the study of plants growth, structure, functions, metabolism,

reproduction, development, and classification. **Example:** studying photosynthesis in leaves.

*** Zoology:** Zoology is the scientific study of animals, including their structures, reproduction, physiology, growth, development, classification and behaviour. **Example:** studying nervous systems of frogs.

*** Microbiology:** Microbiology is the study of microorganisms such as bacteria, amoeba, fungi, protists and viruses etc.

Examples: studying the role of Bacteria in yogurt-fermentation.



Q.2(ii)

A.2(ii)

Interdisciplinary research

collaboration :- Interdisciplinary research

collaboration refers to the practice of combining biology with other scientific disciplines to solve complex problems and advances knowledge. Biology does not work in isolation. It overlaps with physics, chemistry, mathematics, computer science, geography, economics to create new fields of study.

Two examples of interdisciplinary research collaboration :-

Biophysics :- This is the collaboration between physics and biology. Biophysicist

uses the principle of physics (such as energy, motion, forces) to understand biological processes. For examples, understanding how muscles generate force, how nerve impulses travel or how X-ray diffraction was used to determine the DNA's structure.

Biochemistry: This is the collaboration between biology and chemistry. Biochemists study the chemical reactions and molecules within living organisms. For examples: understanding the ^{chemical} structure of ~~proteins~~ proteins and nucleic acids.

Q: 2 (iii) (or)

Ar: 2 (ii) (or)

four kingdom of domain eukarya:-

Kingdom protista

Kingdom Animalia

Kingdom Plantae

Kingdom Fungi

Protista:- Protists include ~~multicellular~~ eukaryotic organisms with unicellular or colonial organization.

Animalia:- Animalia are multicellular eukaryotic organisms. Animals lack chlorophyll and cell wall.

Plantae:- The members of plantae are eukaryotic multicellular organism and contains chlorophyll.

Fungi:- Fungi have eukaryotic organisms which have chitin in their cell wall.

Q.2 (iv) (or)

A:2 (iv) (or)

Prokaryotic cells

absence of nucleus:-

There is an absence of nucleus. Genetic material lies in an area called nucleoid without a nuclear membrane.

membrane-bound

organelles:- There is an absence of mitochondria, chloroplasts, golgi apparatus.

Eukaryotic cells

presence of nucleus:-

There is presence of nucleus true, as well defined nucleus enclosed by a double nuclear envelope with nuclear pores.

membrane-bound

organelles:- There is presence of mitochondria, ER (endoplasmic reticulum), chloroplasts, golgi apparatus.

Size:

Prokaryotic cells are smaller (1-10 μm)

Similar organization, They have complex DNA is also

circular. Ribosomes

are 70S (smaller)

Examples: Bacteria, Archaea

Size:

Eukaryotic cells

are larger (1-10 μm)

They have complex organization. DNA

is linear. Ribosomes

are 80S (larger).

Examples: Plant cell, Animal cell.

Q: 2 (✓)

A: 2 (✓)

Mesophyll cell

Epidermal cells.

Location

These are found in the inner tissue of leaves, between the upper and lower epidermis

These are found on the outer surface of leaves, stems and roots. forming

the outermost protective layer.

Chloroplasts

These contains many chloroplasts. This is main site of photosynthesis in leaves.	These lack ^{chloroplasts} chloroplasts (transparent to allow light to pass through).
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function

Primary function is photosynthesis, they manufacture food (glucose) using light energy, CO_2 , and water.	Primary function is protection, they protect the plant from water loss, mechanical injury and pathogen entry.
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Q: 2 (vi) (or)

A: 2 (vi) (or)

Tissue:- A tissue is a group of similar cells working together to perform a specific function. Cells in tissues have the same origin and similar structure.

four types of tissues:-

⇒ **Epithelial tissue:-** It covers body surfaces and lines body cavities and organs.

function:- Protection, absorption, secretion, filtration.

Examples:- Skin.

⇒ **Muscle tissue:-** It is made up of elongated cells (fibers) containing actin and myosin proteins.

functions: Contraction and movement

types of muscles tissues are:-

1. Smooth muscles

2. Skeletal muscles

3. Cardiac muscles.

p ⇒ Nerve tissue :- It is made up of neurons (nerve cells) with axons and dendrites.

functions :- It transmits electrical impulses (nerve signals) throughout the body for communication.

p ⇒ Connective tissue :- It connects, support & supports, binds with other tissues and organs.

function :- Support, protection and transport.

FOR EXAMPLE :- bone, blood etc.

Q2 (vii)

A: 2 (vii)

Nucleic acids:- Nucleic acid are the most important organic bio-molecules that stores and transfers hereditary information from one generation to next. There are two types of nucleic acids :-

* Deoxyribonucleic acid (DNA)

* ribonucleic acid (RNA)

Both types of nucleic acids (DNA and RNA) are made up of many units called nucleotides.

Purines

Pyrimidines

ring structure

Purines which contains double single ring structure

Pyrimidines which contains single ring structure double ring structure

Size

These are larger in size due to double ring structure

These are smaller in size due to single ring structure

∴ Examples:-

Adenine (A) and guanine (G) are found in both DNA and RNA

Cytosine (C) found in both DNA and RNA, Thymine (T) is only found in DNA and uracil (U) found in RNA.



Q:2 (viii)

A:2 (viii)

Monosaccharides	Polysaccharides	Disaccharides.
Definition	Definition	Definition
Simplest sugar, these cannot be hydrolyzed into simpler sugar.	formed by joining two monosaccharides units through glycosidic bonds with loss of water molecule.	Complex carbohydrate made of many monosaccharides units linked together. They are polymers.
Formula	formula	formula
$C_6H_{12}O_6$	$C_{12}H_{22}O_{11}$	$(C_6H_{10}O_5)_n$
Example	Example	Example
Glucose (blood sugar), fructose fructose (fruit sugar).	Lactose (glucose + galactose), maltose (glucose + glucose)	Starch (energy storage in plants). Cellulose (cell wall).

Q: 2 (ix)

A: 2 (ix)

The four careers are :

* Biotechnology

* Fisheries

* Bio Medicine or surgery

* Horticulture

Biotechnology :-

Biotechnology involves use of living organisms and biological process to make useful products for mankind.

Biotechnologists work in fields include bacteriology, virology and molecular genetics to create medicines, vaccines and genetically modified organisms.

Medicine or Surgery:-

MBBS stands for bachelor of medicine and bachelor of surgery. Medicine is the diagnosis and treatment of different diseases while surgery is the branch which treats the diseases by removal or replacement of the defective parts or organs. After MBBS, a student can specialize in various fields of medicine and surgery.

Q: $2(x) \text{ (or)}$

A: $2(x) \text{ (or)}$

Data:- A collection of facts, observations, and information gathered through experimentation or observations.

Data is typically organized and

and presented in the form of tables, graphs or charts for analysis and interpretation.

Hypothesis:- A tentative answer or proposed explanation to a scientific question. It is based on past experience and prior knowledge. It must be general statement, it must be testable through ~~explain~~ experimentation, it must be falsifiable.

Vector:- A transmitter organisms that carries a parasite or a pathogen from ~~one~~ one organism to another ~~without~~ without being affected itself.

Q: 2 (xi) (or)

A: 2 (xi) (or)

Carolus Linnaeus introduced a naming system to give each organism a name consisting of two latin words.

~~The first name has a capital letter~~

~~and is named~~ The first name is

genus name and the second name is

species. The genus name begins with a

capital & letter and species name begins

with a small letter. It is known

as binomial nomenclature.

Scientific names:-

Humans - Homo sapiens

Potato - Solanum tuberosum.

Section - C

Q: 3 (or)

A: 3 (or)

By adopting the steps of biological method, it was proved that malaria is caused by Plasmodium.

Recognition of problem:-

Malaria was a problem since ancient times but its cause was not known.

Observation:- In 19th century, many different causes of malaria were being suggested. By that time, ~~they~~ there were four major observations about malaria:-

* Malaria and marshy area have some relation

* Drinking the water of marshes does not cause malaria.

* Quinine is an effective drug for treating malaria.

* Plasmodium is seen in the blood of ~~most~~ the malarial patient.

Hypothesis:- Based on these observations and other information, following hypothesis was formulated by the French physician Laveran in 1882.

"Plasmodium is the cause of malaria"

Deduction:- Although hypothesis is a tentative idea. To draw deductions. It is accepted to be true. One of the deductions from the above hypothesis was:-

"If Plasmodium is the cause of malaria then all persons ill with malaria should have Plasmodium"

"in their blood."

Experiments :- This deduction was tested through experiments. Experiment designated as ;

Blood of 100 patients was examined under microscopic microscope. For this example for this purpose of having control group. the blood of 100 healthy people were also examined under microscope.

Result :- The results of experiments showed ~~was~~ that almost all ~~p~~malaria patients had Plasmodium in their blood. Only 7 out of 100 healthy people had plasmodium in their blood. Other 93 healthy persons were without any trace of Plasmodium

in their blood. Plasmodium was in incubation period. The incubation period is time between the entry of parasite in the host and the appearance of the symptoms of disease. After few days those 7 healthy people became malarial patients.

Results were quite convincing to prove the hypothesis that.

"Plasmodium is the cause of malaria"

Reporting the result :-

Result were of these experiments were announced worldwide which helped to control malaria.

Q:4 (or)

#:4 (or)

Proteins :-

Proteins are the main structural or functional components of the cell, which made up of 55% of dry weight of the cells. They are present in all types of cells and contribute in the variety of cell functioning. Proteins are synthesized in the cells by ribosomes.

Chemical composition for proteins :-

Proteins molecules mainly composed of C, H, O, N and sometimes P, S, Fe etc. Chemically proteins are polypeptide chain which are polymers of amino acids.

Amino acids - Amino acids are the building blocks or monomers of proteins. There are about 20-25% different types of amino acids commonly found in proteins. Each amino acid contains a central carbon atom called α (alpha) carbon. Four different groups are attached to amino acids.

- Hydrogen group ($-H$)
- Amino group ($-NH_2$)
- Carboxyl group ($-COOH$)
- An alkyl group ($-R$) also known as functioning group.

Each amino acid has different $-R$ group which determines its chemical properties.

Structure of proteins :-

Structure of proteins may consist of a single polypeptide or more than one polypeptide. Number of amino acids and polypeptide chain can varies in different proteins. Bonds between two amino acids are called peptide bonds.

The structure of protein is classified as primary, secondary, Tertiary or in some cases Quaternary.

Primary structure of proteins:-

Linear chain of amino acids with specific number, type and sequence.

Secondary structure of proteins:-

Upon folding of polypeptide with extra bonds.

Tertiary structure of proteins:-

further foldings of secondary structure with extra bonds.

or go globular form.

Quaternary structure of proteins:-

Two or more polypeptide with tertiary folding further bonded with each other.

Functions of proteins:-

* Antibodies:-

Antibodies protein produced by white blood cells provide immunity against germ.

* Histone:-

It is attached to DNA to form chromosomes.

Enzymes :- Most of all enzymes are proteins which speeds up biochemical reaction.

Q:6 (or)

A:6 (or)

Types and functions of RNA:-

*** Messenger RNA (mRNA) :-**

It carries message of DNA from nucleus to cytoplasm for protein synthesis so called mRNA.

*** Transfer RNA (tRNA) :-**

It transfer amino acids to ribosomes during proteins synthesis, hence called tRNA.

*** Ribosomal RNA (rRNA) :-**

It is the component of RNA therefore

called rRNA. It combines with ribosomal proteins to form units of ribosomes which are factory of proteins.

Process of central dogma :-

DNA (genes) contains information needed for the synthesis of proteins. The flow of biological information, From DNA to RNA and RNA from RNA to proteins is known as the central dogma of molecular biology or central dogma of life.

Proteins are main source structural and functional molecules of living organisms. Correct Primary structure of any proteins is pri-

mainly determined ^{by} ~~to~~ DNA. Protein synthesis takes place in two sequential steps transcription and translation. Together translation and transcription are known as gene expression.

Transcription:

Transcription is the first step of gene expression that involves the formation of RNA molecule from DNA. Both DNA strands of a gene unwind from each other. Only one strand of DNA is copied. Nucleotide sequence of one DNA strand (coding strand) is copied in form of mRNA. After the gene is copied to mRNA, both strands ~~are~~

DNA are again copied together to its normal double stranded shape.

Translation:

The mRNA synthesized in the nucleus of the cell moves into the cytoplasm where it binds to the ribosome. The mRNA message (codes) is translated by ribosomes which link amino acids to form polypeptide chain with specific sequence of amino acids. Different types of tRNA transfers amino acids to ribosome as per sequence of code in mRNA.